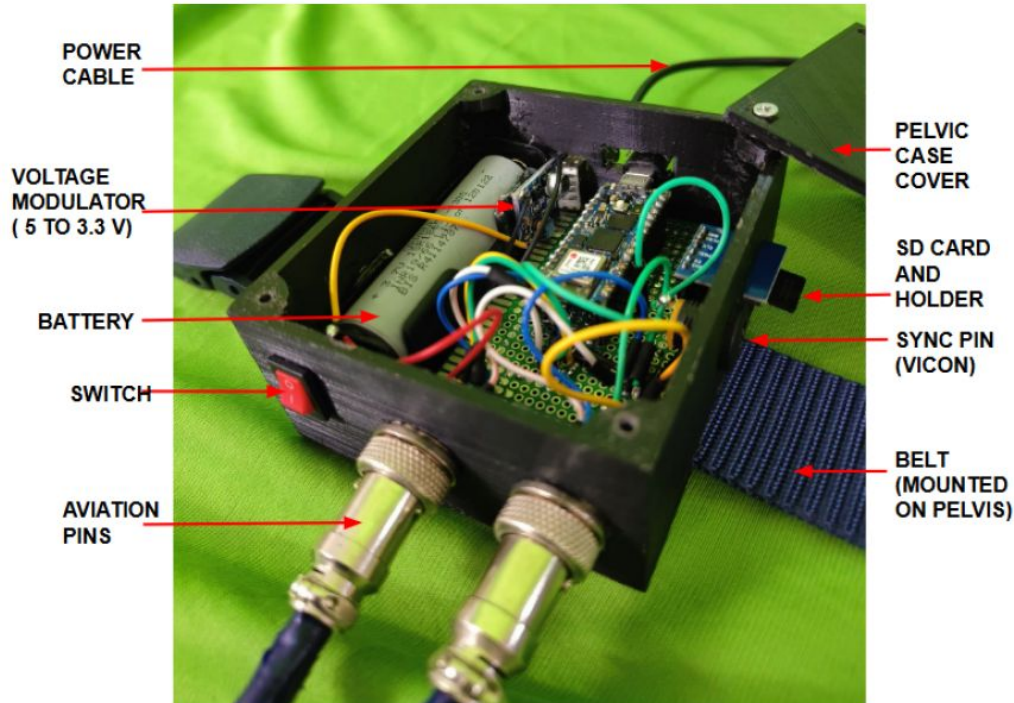
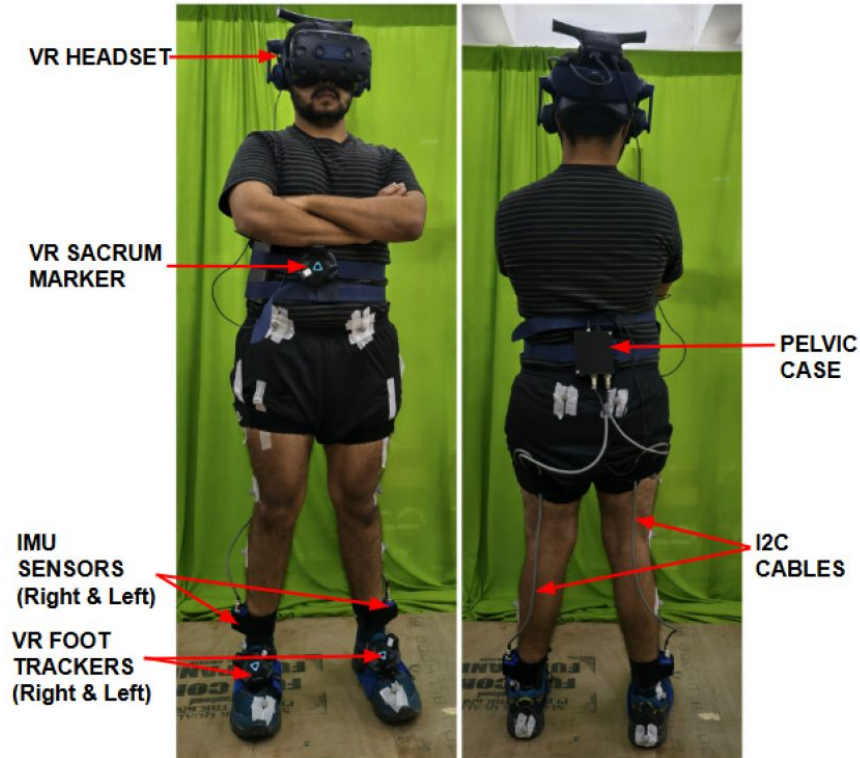


Hardware Prototype: Wearable IMU System (50 Hz)

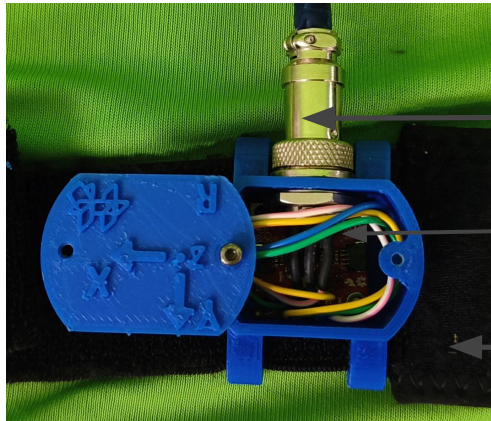
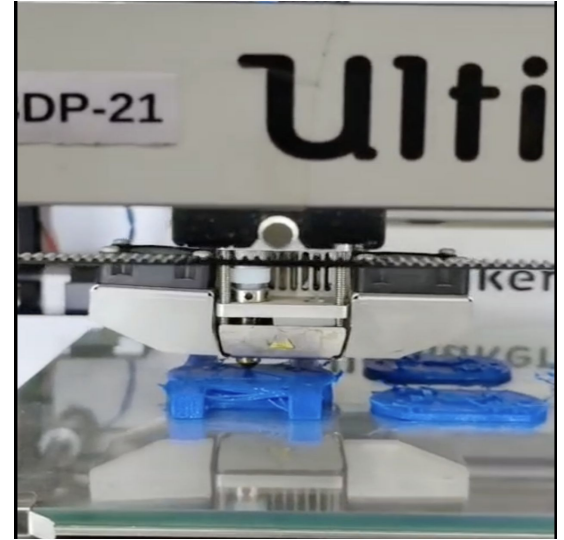
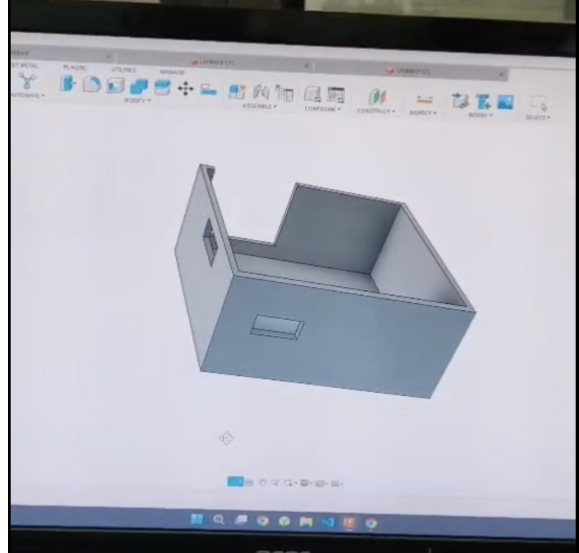
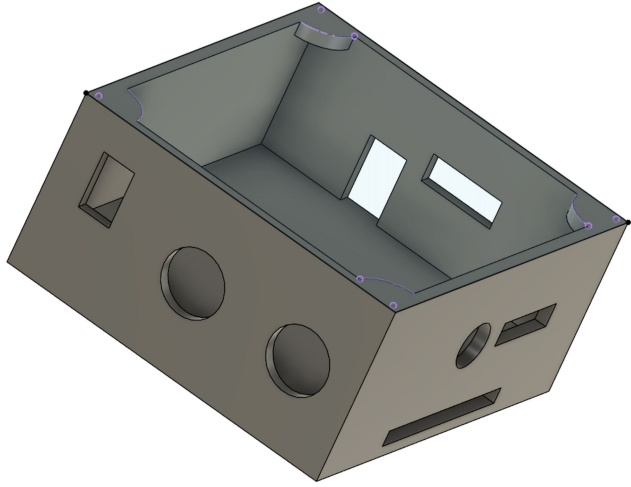
(a) Pelvic module: Arduino Nano **RP2040** microcontroller
| Sparkfun **ICM-20948** | 3.7 V Lithium-ion | I2C wiring (4
core) - SDA(data), SCL(clock), ground and voltage in.



(b) Full-body IMU, VR and Vicon marker setup



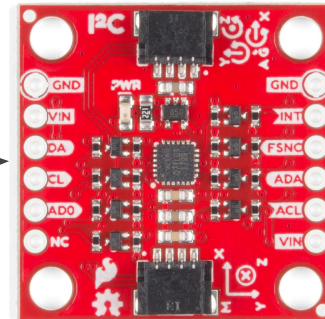
CAD DESIGN (Autodesk Fusion 360)



AVIATION PINS

IMU

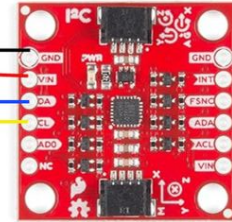
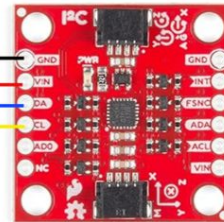
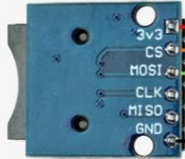
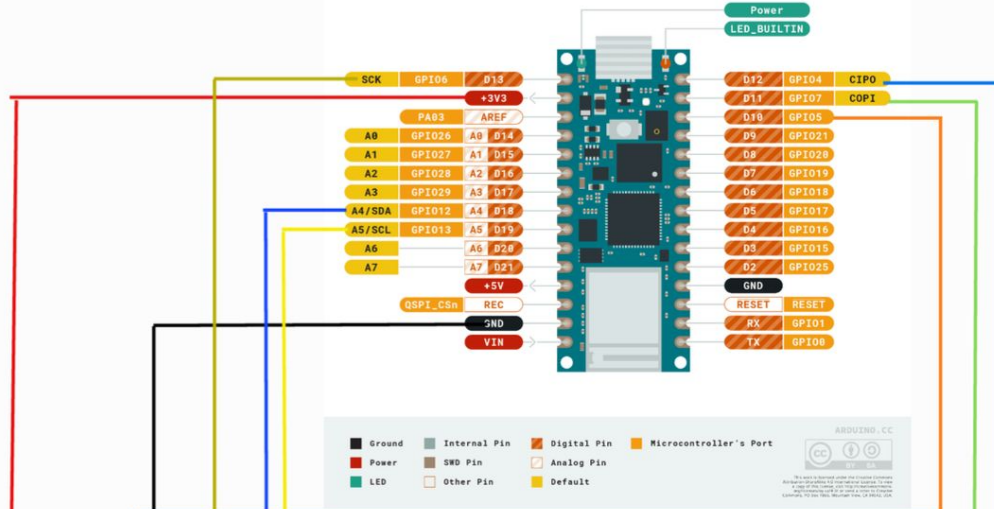
BELT (MOUNTED ON SHANK)



9 DOF:
3 axis accelerometer
3 axis gyroscope
3 axis magnetometer

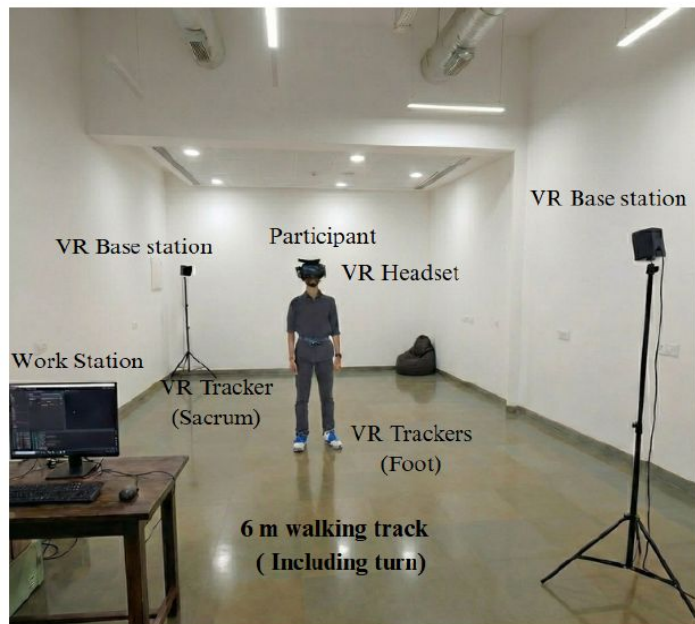


ARDUINO NANO RP2040 CONNECT

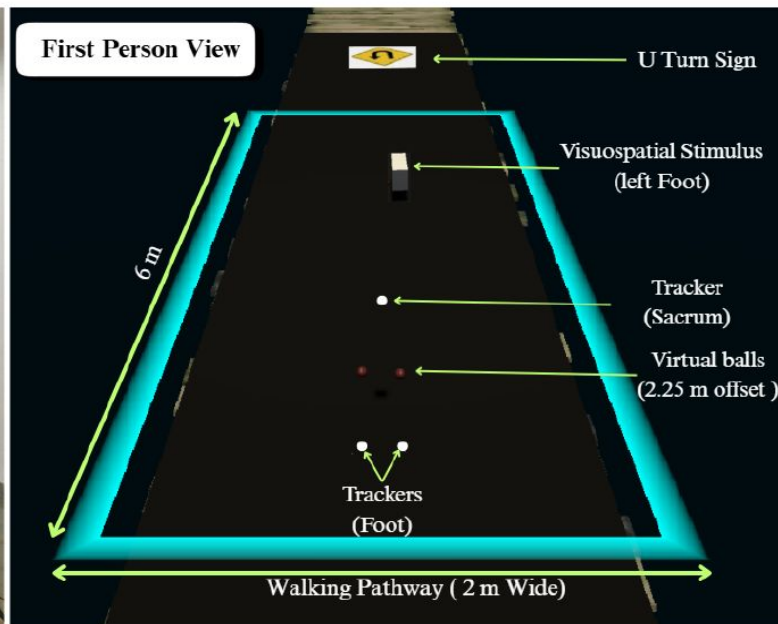


Hardware Prototype: VR Setup (90 Hz)

- Virtual bridge environment: to simulate a constrained walking pathway
- Walking trials: **Baseline(no perturbations)** and **perturbation trials** [White box: left, Green box: Right]
- Trackers: real-time positional data for reconstructing motion



(a) VR Sensor Setup



(b) Experimental Setup

VR Environment and Unity post processing codes credits: Chetan Sharma(HCR Lab)

Hardware Prototype: Vicon (Plug-in-Gait Model) (200 Hz)

- CALIBRATION
- 8 infrared cameras: track reflective markers
- High-precision 3D kinematic data: **gold standard**
- Accurate detection of gait events, allowing computation of spatiotemporal gait parameters.

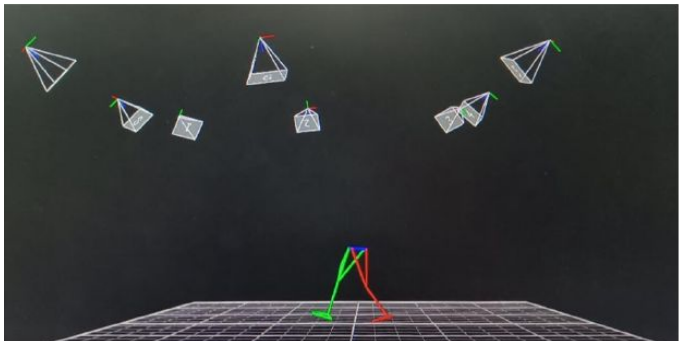
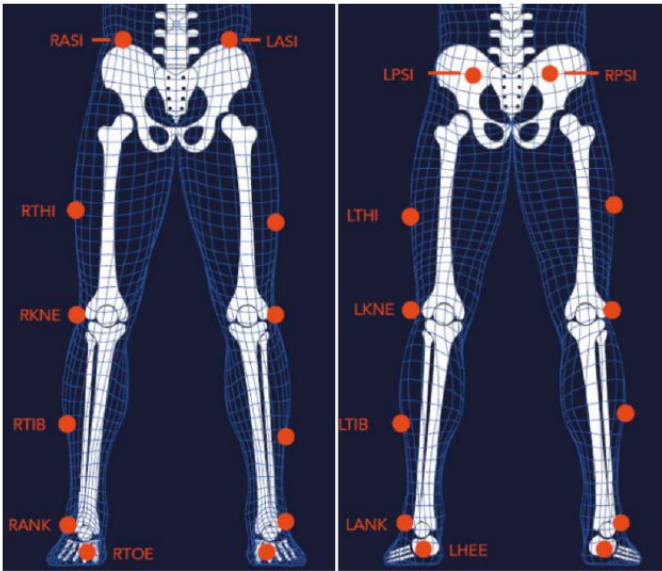


Table 2.3: Plug-in Gait lower-body marker configuration

Region	Right Marker	Left Marker	Anatomical Location
Pelvis	RASI	LASI	Anterior Superior Iliac Spine (ASIS)
Pelvis	RPSI	LPSI	Posterior Superior Iliac Spine (PSIS)
Thigh	RTHI	LTHI	Mid-thigh, lateral surface
Knee	RKNE	LKNE	Lateral femoral epicondyle
Shank	RTIB	LTIB	Mid-shank, lateral surface
Ankle	RANK	LANK	Lateral malleolus
Heel	RHEE	LHEE	Calcaneus (heel)
Toe	RTOE	LTOE	Second metatarsal head (toe)



(a) Plug-in Gait marker placement (front and back)